

Mark schemes

Q1.

(a) $A = 80$

B1

1

(b) $5000 = 80 \times k^{56}$

SC1 Verification. Need 62.51 or better

M1

$$k = \sqrt[56]{\frac{5000}{80}} \approx 1.07664$$

Or using logs: M1 $\ln \left(\frac{5000}{80} \right) = 56 \ln k$

A1 $k = e^{\ln \left(\frac{62.5}{56} \right)}$

Or 3/3 for $k = 1.076636$

Or 1.076637 seen

M1A1

3

(c) (i) $V = 80 \times k^{106} = 200707$

200648 using full register k

M1A1

2

(ii) $\ln 10000 = \ln k^t$

M1

$$t = \frac{\ln 10000}{\ln k} = 124.7 \Rightarrow 2024$$

M1 $t \ln k = \ln 10000$

A1 CAO

Or trial and improvement M1 expression

M1 125, 124, A1 2024

M1A1

3

[9]

Q2.

(a) $\log_a x = \log_a 6^2 - \log_a 3$

One law of logs used correctly

M1

$$\log_a x = \log_a \left(\frac{6^2}{3} \right)$$

A second law of logs used correctly

M1

$$\log_a x = \log_a \frac{36}{3} \Rightarrow x = 12$$

CSO AG

A1

3

(b) $\log_a y + \log_a 5 = 7 \Rightarrow \log_a 5y = 7$

M1

$$\Rightarrow 5y = a^7 \text{ or } y = \frac{1}{5} a^7 \text{ or } a = (5y)^{1/7}$$

Eliminates logs

m1

Accept these forms

A1

3

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[6]

Q3.

(a) (i) y -coordinate of A is $27 - 3^0 = 26$

M1A1

2

(ii) When $x = 3$, $y = 27 - 3^3 = 0 \Rightarrow B(3,0)$

AG; be convinced

B1

1

(b) $h = 1$

PI

B1

Area $\approx h/2\{\dots\}$
 $\{\dots\} = f(0) + f(3) + 2[f(1) + f(2)]$
OE summing of areas of the 'trapezia'..

M1

$\{\dots\} = "26" + 0 + 2(24 + 18)$
on (a)(i) ($\Sigma trap = "25" + 21 + 9$)

A1 ✓

(Area $\approx$) 55
on $[42 + 0.5 \times "(a)(i)"]$

A1 ✓

4

(c) (i) $\log_{10} 3^x = \log_{10} 13$
Takes $\ln$ or $\log_{10}$ on both or $x = \log_3 13$

M1

$$x \log_{10} 3 = \log_{10} 13$$

Use of $\log 3^x = x \log 3$ or $\log_3 13 = \frac{\lg 13}{\lg 3}$ OE
(PI by $\log_3 13 = 2.335$ or better)

m1

$$x = \frac{\lg 13}{\lg 3} = 2.334717 \dots = 2.3347 \text{ to 4dp}$$

Must show that logarithms have been used

A1

3

(ii) $\{k\} = 14$

Condone $y = 14$; Accept final answer 14 with only zeros after decimal point e.g. 14.000

B1

1

(d) (i) Translation;
'Translation' / 'translate(d)'
B0 if more than one transformation

B1

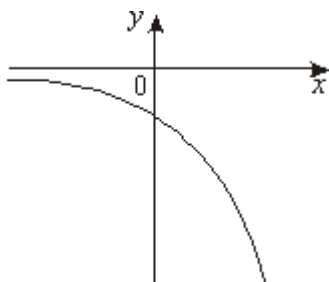
$$\begin{bmatrix} 0 \\ -27 \end{bmatrix}$$

Accept full equivalent in words provided
 linked to 'translation/move/shift' and negative y-direction
 (Note: B0 B1 is possible)

B1

2

(ii)



Correct shape (translation of given curve vertically downwards)

B1

Only point of intersection with coordinate axes is on negative y-axis and curve is asymptotic to the negative x-axis

B1

2

[15]

Q4.

(a) $y = e^{2x} - 10e^x + 12x$

(i) $\frac{dy}{dx} = 2e^{2x} - 10e^x + 12$
 $2e^{2x}$

B1

remaining terms correct, no extras

B1

2

(ii) $\frac{d^2y}{dx^2} = 4e^{2x} - 10e^x$
 ft 1 slip

B1F

(b) (i) $2e^{2x} - 10e^x + 12 = 0$
 $e^{2x} - 5e^x + 6 = 0$

AG (be convinced)

B1

(ii) $z^2 - 5z + 6 = 0$

use of $z = e^x$ oe

M1

$z = 2, 3$

$z = 2, e^x = 2$

$x = \ln 2$

finding $e^x =$ their 2,3

M1

$z = 3, e^x = 3$

$x = \ln 3$

all correct AG

A1

SC: verification

$\ln 2$ (B1)

$\ln 3$ (B1)

(iii) $x = \ln 2:$

$y = e^{2\ln 2} - 10e^{\ln 2} + 12\ln 2$

either substitution of their $x = \ln 2$ ($e^x = 2$)

or their $x = \ln 3$ ($e^x = 3$)

M1

or

$2^2 - 10 \times 2 + 12 \ln 2 = 4 - 20 + 12 \ln 2 = -16 + 12 \ln 2$

A1

$x = \ln 3$.

$y = e^{2\ln 3} - 10e^{\ln 3} + 12\ln 3 = 9 - 30 + 12 \ln 3 = -21 + 12 \ln 3$

A1

3

(iv) $x = \ln 2$:

$$\frac{d^2y}{dx^2} = 4e^{2\ln 2} - 10e^{\ln 2}$$

use of; in either of their $e^x = 2, 3$ into their $\frac{d^2y}{dx^2}$

M1

$$= 16 - 20 = -4$$

$\therefore$ maximum

CSO

A1

$x = \ln 3$:

$$\frac{d^2y}{dx^2} = 4e^{2\ln 3} - 10e^{\ln 3}$$

$$= 36 - 30 = 6$$

$\therefore$ minimum

CSO

A1

3

[13]

Q5.

(a) $f(x) = 2e^{3x} - 1$

Range: $f(x) > -1$ (or $y > -1$ or $f > -1$)
for -1 only

M1

exactly correct

A1

2

(b) $y = 2e^{3x} - 1$

$$x = 2e^{3y} - 1$$

$$x \leftrightarrow y$$

M1

$$2e^{3y} = x + 1$$

$$e^{3y} = \frac{x+1}{2}$$

attempt to isolate

M1

$$y = \frac{1}{3} \ln \left(\frac{x+1}{2} \right)$$

all correct with no error AG (be convinced)

A1

3

(c) $f^{-1}(x) = \frac{1}{3} \left(\frac{2}{x+1} \right) \times \frac{1}{2} \quad \text{OE}$

for differentiation of $\ln$; $\frac{k}{\text{their}(x \pm 1)}$

M1

for $\frac{1}{2}$

A1

all correct

A1

$$x = 0$$

$$f^{-1}(x) = \frac{1}{3}$$

CSO

A1

4

Alternative

$$f^{-1}(x) = \frac{1}{3} \ln(x+1) - \frac{1}{3} \ln 2$$

M1A1

$$f^{-1}(x) = \frac{1}{3(x+1)}$$

A1

$$f^{-1}(0) = \frac{1}{3}$$

CSO

A1

[9]

Q6.

- (a) (i) $(5000 - x)$ seen in a product
Could be implied, eg $5000a - xa$

B1

$$\frac{dx}{dt} = kx(5000 - x)$$

B1

2

- (ii) $200 = k \times 1000 \times (5000 - 1000)$
 $\frac{dx}{dt} = 200, x = 1000$ in their diff. equation
Condone ts and $t = 0$ for M1

M1

$$k = 0.00005$$

CAO OE

A1

2

- (b) (i) $t = 4 \ln \left(\frac{4 \times 2500}{5000 - 2500} \right) = 5.5$ (hours)
 $x \rightarrow 2500$ (or $4 \ln 4$)

M1

CAO

A1

2

(ii) $e^{\frac{30}{4}}$
 $e^{7.5} = \frac{4x}{5000 - x}$

B1

OE

M1

$$5000 \times e^{7.5} = x(4 + e^{7.5})$$

Soluble for x

m1

$$x = 4988.96.. \Rightarrow 4989 \text{ rabbits infected}$$

Or 4988 or 4990; integer value only

A1

4

[10]

Q7.

(a) $\log_a x = \log_a 6^3 - \log_a 8$

A law of logs used correctly

M1

$$\log_a x = \log_a (6^3 \div 8)$$

A different law of logs used correctly

M1

$$x = 6^3 \div 8 = 27$$

CSO AG (be convinced)

A1

3

ALT $\log_a x = 3 \log_a 6 - 3 \log_a 2$

(M1)

$$\frac{1}{3} \log_a x = \log_a \frac{6}{2}$$

(M1)

$$x^{\frac{1}{3}} = 3 \Rightarrow x = 27$$

(A1) CSO

(b) (i) $\log_4 1 = 0$

B1

(ii) $\log_4 4 = 1$

SC in (b): For all four answers $\frac{1}{4}$; 1; $\frac{1}{2}$; 2

B1

- (iii) $\log_4 2 = 0.5$
give 0/4; otherwise mark each

B1

- (iv) $\log_4 8 = 1.5$
independently

B1

4

[7]

Q8.

- (a) {y-coordinate of A is} 2

B1

1

- (b) (i) $h = 0.25$

B1

$$\text{Integral} = \frac{1}{2} \{ \dots \}$$

$$\{ \dots \} =$$

$$f(0) + 2\left[f\left(\frac{1}{4}\right) + f\left(\frac{1}{2}\right) + f\left(\frac{3}{4}\right)\right] + f(1)$$

$$\{ \dots \} =$$

M1

$$2 + 4 + 2[(2.316.. + 2.732.. + 3.279(5..)]$$

A1 ✓

$$\{ = 6 + 2 \times 8.3276.. \} \{ = 22.65(5...) \}$$

Condone one numerical slip
Accept values to 3sf (rnd or trunc)
ft answer from (a) if not "2"

$$\text{Integral} = 0.125 \times 22.655.. = 2.8319..$$

$$\text{Integral} = 2.83 \text{ to 3 sf}$$

CAO Must be 2.83
(NMS scores 0/4)

A1

4

- (ii) Relevant trapezia drawn on a copy of given graph
Accept relevant single trapezium with its sloping side above the curve

M1

{Approximation is an}overestimate

A1

2

(c) $5 = 3^x + 1 \Rightarrow 3^x = 4$

B1

$$\log_{10} 3^x = \log_{10} 4$$

Takes ln or $\log_{10}$ on both sides of $3^x = k$, where $k > 0$

M1

$$x \log_{10} 3 = \log_{10} 4$$

Use of $\log 3^x = x \log 3$

m1

$$x = \frac{\lg 4}{\lg 3} = 1.26185 \dots = 1.2619 \text{ to 4dp}$$

*Accept 4dp or better
 [If using T&I a full justification is required; else M0m0A0]*

A1

4

(d) $f(x) = 3^{-x} + 1$

B1

1

[12]

Q9.

(a) $2e^x = 5$

$$e^x = \frac{5}{2}$$

(exact)

M1

$$x = \ln \frac{5}{2} \quad (0.916)$$

(A0 if further wrong work)

A1

2

(b) (i) $2e^x + 5e^{-x} = 7$
 $2e^{2x} + 5 = 7e^x$
Dealing with e^{-x}

M1

$$2y^2 - 7y + 5 = 0$$

AG

A1

2

(ii) $(2y - 5)(y - 1) = 0$

attempt to solve $y = \frac{5}{2}, 1$ (SC B1)

M1

$$x = \ln \frac{5}{2}$$

$$e^x = \frac{5}{2}$$

A1

$$x = 0 \text{ (or } \ln 1)$$

$$e^x = 1$$

A1

3

[7]